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KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI, GHANA

AE 351

Soil and Water Engineering I

Lecture 3

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INFILTRATION, PERCOLATION AND GROUNDWATER HYDROLOGY

- Rainfall-infiltration is a key link in the process of water transformation of various ecosystems.
- Through infiltration, rainfall is turned into soil water and used by crops.
- Simultaneously, the change of rainfall infiltration affects the development of soil and water.
- Infiltration is the downward entry of water into the soil from the soil surface. Percolation on the other hand is the downward movement of water through saturated or nearly saturated soil in response to the force of gravity.

Effects of Infiltration/Percolation

- Percolation occurs when water is under pressure or when the tension is smaller than about ½ atmospheres. Percolation rate is synonymous with infiltration rate with the qualitative provision of saturated or near saturated conditions.

- Slow water penetration or slow infiltration affects crop production. The rate of the infiltration process relative to the rate of water supply determines how much water will enter the root zone, and how much, if any, will run-off. Infiltrating water replenishes soil moisture deficit and thereafter moves on downward to charge groundwater. Where infiltration rate (IR) is restricted:
- Plants may be denied sufficient moisture
- While the amount of erosion increases.

Definition and Characteristics of Infiltration

- Infiltration rate is defined as the volume flux of water flowing into the soil profile per unit surface area of soil, and this flux has the dimension of velocity.
- Infiltration capacity is the maximum rate at which rain can be absorbed by a soil in a given condition.
- Soil infiltration capacity is initially high, particularly when the soil is initially dry, but it tends to decrease monotonically until it asymptotically approaches a constant rate, which is referred to as the final, basic, primary, terminal or steady state infiltration rate.

- When the delivery rate of water exceeds soil infiltration capacity, it is the latter which determines the actual IR, and the process becomes surface controlled. For soils with stable surfaces, decreases in infiltration capacity result from the inevitable decrease in the matric suction gradient which occurs as infiltration proceeds. In the initial wetting process, the surface is saturated but the soil below is dry and the matric suction gradient is very steep. As the wetted zone deepens, this gradient is gradually reduced. Decrease in infiltration capacity can also result from gradual deterioration of the soils structure and the formation of a surface seal or crust.

Factors Affecting Infiltration Capacity:

- Soil moisture
- Compaction due to rain
- In-wash of fine soil material
- Compaction due to animals/humans
- Macrostructure of soil
- Vegetation cover
- Entrapped air
- Temperature
- Surface detention and thickness of saturated layer and
- The effects of freezing.

Models of Infiltration

There are many models but the classical **Hortonian** infiltration model is represented as:

- $i = i_c + (i_o - i_c)\, e^{-kt}$

Where:

- **i** = Infiltration rate
- i_c = Asymptotic steady state infiltration rate reached when the time, t and the cumulative volume of water infiltrated become large
- i_o = Initial infiltration capacity of the soil
- **k** = Constant which determines how quickly i will decrease from i_o to i_c.

Measurement of Infiltration Rate

- Blocked furrow infiltrometers
- Cylinder infiltrometers
- Inflow-outflow measurement technique
- Volume balance equation.

❖Cylinders are preferable because they are simple, inexpensive and portable.

Limitations of Ring Infiltrometers

- Disturbance of soil during emplacement
- Lateral flow of water underneath cylinders
- Water seepage down the cylinder-soil interface
- Air within and underneath the cylinder
- Effect of depth of cylinder emplacement
- Effect of head of water over the soil
- Effect of temperature of the soil and water
- Representation of the effects of rain beat

These Limitations Can Be Minimised By

- Limiting the insertion of cylinder to a shallow depth of about 50 mm
- Minimising soil disturbance
- Height of water in cylinders must ideally be <50 mm
- Use **'buffer zone'** (double ring) to reduce lateral flow and enhance accuracy of the vertical infiltration measurement.
- **NB:**
  - The inner diameter of the inner ring is 30 cm and that of the outer ring is 60 cm.
  - The height of the rings is 50 cm.
  - Water levels in the inner ring can be measured at 1, 5, 10, 20, 30, 45, 60, 90 and 120 minutes or more frequently if infiltration is rapid.
  - Refill the cylinder when it drops to 10 cm from the initial water height of 15 cm.
  - Water levels in the inner and outer rings should be maintained at the same level.
  - Infiltration tests should be continued for 1-4 h until steady state is reached depending on the soil type. The lower and higher limits for clayey and sandy soils respectively.

General Hydrologic Budget

- $F = B_I + P - E - T - ET - S - R - I_A - B_O$

Where

- F is infiltration, which can be measured as a volume or length;
- B_I is the boundary input, which is essentially the output watershed from adjacent, directly connected impervious areas;
- B_O is the boundary output, which is also related to surface runoff, R, depending on where one chooses to define the exit point or points for the boundary output;
- P is precipitation;
- E is evaporation;
- ET is evapotranspiration;
- S is the storage through either retention or detention areas;
- I_A is the initial abstraction, which is the short term surface storage such as puddles or even possibly detention ponds depending on size;
- R is surface runoff.

Groundwater in the Soil

- In the vadose zone, soil pores contain either water or air. Therefore it is referred to as zone of aeration.

- In the phreatic zone, interstices are filled with water and are known as the zone of saturation. This zone might extend to considerable depths. But as depth increases the weight of overburden tends to close pore spaces and relatively little water is found at depths greater than 3 km.

- Groundwater is present in units of rock or formations. Therefore for water to be present there must be porosity i.e.
- Inter-granular or primary porosity and
- Fracture or secondary porosity e.g. fractured sandstones

Aquifer and Properties

- **Porosity = Volume of voids/Bulk volume of rock**
- An aquifer is a geological formation, group of formations, or part of a formation that contains sufficient saturated permeable material to yield significant quantities of water to wells and springs. It has porosity, permeability and is in a state of moisture saturation. Confined aquifers are sometimes called artesian aquifers.
- Confined aquifer has pressure that can cause water to rise above the upper aquifer confining layer which have very low permeability since in theory there is no such thing as impermeable rock. In a confined or water table aquifer, the upper limit of saturation i.e. the water table is at atmospheric pressure. At any point below the water table, the water is above atmospheric and at any point above the water table the pressure is less than atmospheric.

- **Specific Yield (S_y) =** Ratio of the volume of water that will drain by gravity from a rock or soil that was initially saturated (V_w) to the bulk volume (V_b)

- **Specific Retention (Sr) =** Volume of water that a rock or soil after being saturated will retain against the pull of gravity (Vr) to the bulk volume (V_b). Specific retention refers to the water that is unable to drain from the pores

- **Rise of watertable =** infiltration/specific yield (expressed as a fraction)

Thank You

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